Abstract

This work presents HALO (*Holographic Artificial-Life with Observational confinement*), a quantum-computing architecture for agent-based artificial life (ALife) that unifies four independently established quantum-information primitives: the Page–Wootters mechanism for emergent relational time, decoherence-free subspace (DFS) encoding for collective noise suppression, subspace-isometric holographic compression, and complexity-theoretic fitness evaluation grounded in local Hamiltonian hardness. Together, these components yield a fully quantum agent model in which temporal evolution, noise protection, state compression, and fitness selection are all realised intrinsically, without classical mid-computation control.

A central methodological contribution is a *claim ladder*: a hierarchy of assertions stratified by evidential strength, each carrying explicit assumptions and a falsifiable evidence threshold. This structure cleanly separates demonstrated results from open conjectures, enabling incremental and independently verifiable progress on a multi-component architecture.

Validated evidence is reported at six levels. *C0:* A five-qubit noiseless circuit confirms the composability of clock-conditioned evolution, reversible ancilla uncomputation, and DFS noise resilience. *C1:* A matched-fidelity lattice-Trotter proxy benchmark establishes sub-linear qubit-count scaling (log-log slope 0.4315 $[0.2823, 0.5715]_{95\%}$) for binary-encoded circuits against a one-hot baseline, with an explicit circuit-depth tradeoff reported. *C2:* Isometric embeddings of k-excitation subspaces ($k = 1, 2, 3$) into $\lceil \log_2 \binom{N}{k} \rceil$ boundary qubits are constructed and verified for bulk sizes $N \in \{4, 8, 16, 32\}$, achieving compression ratios up to $6.40\times$ with zero isometry error across all twelve instances. *C3:* Under a bounded heuristic attacker (100 trials per size, $n = 6$–18), combined attack success on masked parity-Hamiltonian proxies decreases from 0.69 to 0.20; a Mann–Kendall trend test yields $S = -21$, $p = 0.0027$. *C4:* Direct-sum isometric embeddings spanning multiple excitation sectors simultaneously are constructed for sector combinations $K \in \{\{1, 2\}, \{0, 1, 2\}, \{1, 2, 3\}\}$ at bulk sizes $N \in \{4, 6\}$, all achieving zero isometry error; sector leakage under a transverse-field perturbation H_Γ is quantified and shown to be zero at $\Gamma = 0$. *C5:* A fully quantum multi-generation evolution over $G = 5$ generations with $P = 8$ agents—implemented as a Qiskit 5-qubit circuit—demonstrates Grover-style amplitude amplification with no intermediate measurement: fit-agent probability oscillates from 0.25 to 0.82 and back across successive generations, confirming the coherent selection-plus-mutation loop. Universal holographic compression, unconditional cryptographic security, and hardware-scale validation are identified as open problems and explicitly deferred (C6). All results are reproducible from the accompanying artifact package.

A Tiered Evidence Program for a Holographic Quantum Artificial-Life Architecture

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1 Introduction

The intersection of quantum computation and artificial life has received growing theoretical attention [1, 2, 3]. Quantum systems offer two structural features that are absent in classical ALife models: exponentially large Hilbert spaces that can encode structured superpositions of agent configurations, and entanglement that permits genuinely non-local correlations between agents and their environments. Despite this promise, most existing quantum ALife proposals remain at the level of qualitative analogy, lacking both a concrete architectural blueprint and a rigorous experimental validation program.

The present work addresses this gap by introducing Halo, an architecture that draws on four independently well-studied primitives: (i) the Page–Wootters (PW) mechanism [4, 5], which derives an emergent notion of time from the entanglement between a clock subsystem and the rest of the universe, permitting agent evolution without an external classical clock; (ii) decoherence-free subspaces (DFS) [6, 7], which protect logical information from collective environmental noise without active feedback; (iii) subspace-isometric compression inspired by the holographic principle [8, 11], which maps a lower-dimensional boundary register to a structured bulk subspace; and (iv) QMA-hard fitness evaluation [12, 14], which makes the fitness landscape computationally inaccessible to polynomial observers, providing a notion of evolutionary privacy.

1.1 The claim-ladder methodology

A recurring problem in speculative quantum frameworks is the conflation of architectural *possibility* with experimental *evidence*. It is not forbidden by any known physical principle that all four primitives above could be integrated at macroscopic scale; yet no experiment currently demonstrates this. To avoid this conflation, a *claim ladder* is adopted: a hierarchy of claims $C_0 \subset C_1 \subset \cdots \subset C_6$, each strictly stronger than the preceding, with the following structure:

C0 Toy-scale composability of the core primitives on a noiseless simulator.

C1 Relative resource-efficiency improvement under a declared proxy benchmark family, with a matched-fidelity fairness gate.

C2 Explicit isometric compression in a restricted subspace family, verified by algebraic metrics.

C3 Declining attack success under a bounded heuristic attacker on proxy Hamiltonian instances.

C4 Direct-sum isometric compression spanning multiple excitation sectors simultaneously, with leakage characterised under a transverse-field perturbation, compiled as a Qiskit circuit.

C5 Fully quantum multi-generation evolution via Grover amplitude amplification, with coherent mutation and no intermediate measurement, implemented as a Qiskit 5-qubit circuit.

C6 Universal macroscopic holographic compression, unconditional cryptographic security, hardware-scale validation, and practical scalability. *[Deferred.]*

Each level is evaluated separately; progress at one level does not entail the next. This methodology is inspired by similar tiered claim structures in quantum error correction benchmarking [15] and is, to the author's knowledge, the first application of this approach to a quantum ALife proposal.

1.2 Contributions

The main contributions of this paper are:

1. A concrete seven-component Halo architecture integrating PW clocks, DFS encoding, holographic compression, and fitness encryption (section 3).
2. A complete evidence program with declared proxy families, falsifiable thresholds, and reproducible artifacts (section 4).
3. Experimental results at levels C0–C5, including the first matched-fidelity log-qubit scaling benchmark for PW-clock circuits; the first explicit isometry constructions for k-excitation holographic compression ($k = 1, 2, 3$); direct-sum cross-sector isometries with leakage characterisation (C4); and a fully coherent multi-generation quantum evolution loop via Grover amplitude amplification (C5) (section 5).
4. Explicit characterisation of what is *not* demonstrated, with a roadmap of open problems (section 7).

Paper organisation. Section 2 reviews the four theoretical pillars. Section 3 describes the Halo architecture. Section 4 defines the claim ladder and experimental design. Section 5 presents and analyses simulation results. Section 6 discusses resource tradeoffs and related work. Section 7 lists open problems. Section 8 concludes. Reproducibility details appear in appendix A.

2 Background

2.1 Page–Wootters mechanism

In standard quantum mechanics, time enters as an external parameter. The Page–Wootters (PW) mechanism provides an alternative: a static global state $|\Psi\rangle$ satisfying the Wheeler–DeWitt equation $\hat{H}_{\text{total}} |\Psi\rangle = 0$ can yield a non-trivial conditional evolution when a designated *clock* subsystem C is traced or conditioned upon.

Specifically, let the global Hilbert space split as $\mathcal{H} = \mathcal{H}_C \otimes \mathcal{H}_S$, where S is the *system* (here, the agent register). The Wheeler–DeWitt condition reads

$$(H_C \otimes \mathbb{I}_S + \mathbb{I}_C \otimes H_S) |\Psi\rangle = 0. \tag{1}$$

Giovannetti, Lloyd, and Maccone [5] showed that conditional states $|\psi_t\rangle_S = {}_C\langle t| \Psi\rangle$ satisfy the Schrödinger equation in t, recovering standard unitary evolution as an emergent phenomenon. In the discrete implementation adopted here, the clock is a single qubit, and "time" takes values $t \in \{0, 1\}$. This is a minimal proof-of-principle; extensions to multi-qubit clocks with higher temporal resolution are straightforward in principle but incur additional qubit overhead.

2.2 Decoherence-free subspaces

A decoherence-free subspace [6, 7] is a subspace $\mathcal{H}_{\mathrm{DFS}} \subset \mathcal{H}$ that is invariant under a given class of error operators. For collective dephasing—where the environment couples to every qubit with the same Z operator—the relevant condition is that every state in $\mathcal{H}_{\mathrm{DFS}}$ is a $+1$ eigenstate of the total Z symmetry, i.e., $\sum_i Z_i |\psi\rangle = 0$ for all $|\psi\rangle \in \mathcal{H}_{\mathrm{DFS}}$. The two-qubit subspace spanned by $\{|01\rangle, |10\rangle\}$ satisfies this condition and encodes one logical qubit.

Under a collective noise channel $\mathcal{E}(\rho) = e^{-i\theta(Z\otimes\mathbb{I}+\mathbb{I}\otimes Z)/2} \rho\, e^{+i\theta(\cdots)/2}$, both $|01\rangle$ and $|10\rangle$ acquire the same global phase, leaving Born-rule probabilities invariant. DFS encoding does not protect against *individual* qubit errors; this is treated as an acknowledged limitation (see section 6).

2.3 Subspace-isometric holographic compression

The holographic principle, as formalised in AdS/CFT correspondence [8], asserts that a d-dimensional bulk theory has a dual description on a $(d-1)$-dimensional boundary. In a quantum information context, Harlow and Hayden [9] and subsequent work on quantum error-correcting codes in AdS/CFT [10, 11] established that holographic dictionaries can be realised as isometric maps $V : \mathcal{H}_{\mathrm{boundary}} \hookrightarrow \mathcal{H}_{\mathrm{bulk}}$ satisfying $V^\dagger V = \mathbb{I}_{\mathrm{boundary}}$.

The construction is restricted to the tractable case of the single-excitation subspace. For an N-site system, this subspace has dimension N (spanned by $\{|e_j\rangle : j = 1, \ldots, N\}$ where $|e_j\rangle$ has a single excitation at site j). A boundary register of $\lceil \log_2 N \rceil$ qubits can represent this subspace with compression ratio $N/\lceil \log_2 N \rceil$, which grows without bound as $N \to \infty$. The isometry V is then an explicit unitary circuit embedding this N-dimensional space into $2^{\lceil \log_2 N \rceil}$-dimensional boundary Hilbert space. This is a mathematical statement about subspace structure, not a claim about arbitrary Hamiltonian universality.

2.4 QMA-hardness and fitness encryption

QMA (Quantum Merlin-Arthur) is the quantum analogue of NP, containing decision problems whose yes-instances have a polynomial-size quantum proof that can be verified in polynomial quantum time. The local Hamiltonian problem—deciding whether the ground energy of a k-local Hamiltonian is below a threshold—is QMA-complete for $k \geq 2$ [12, 13].

The fitness metric is the ground-state energy of a masked local Hamiltonian chosen from this class. The intent is that evaluating raw fitness (finding the ground energy) is computationally hard for any

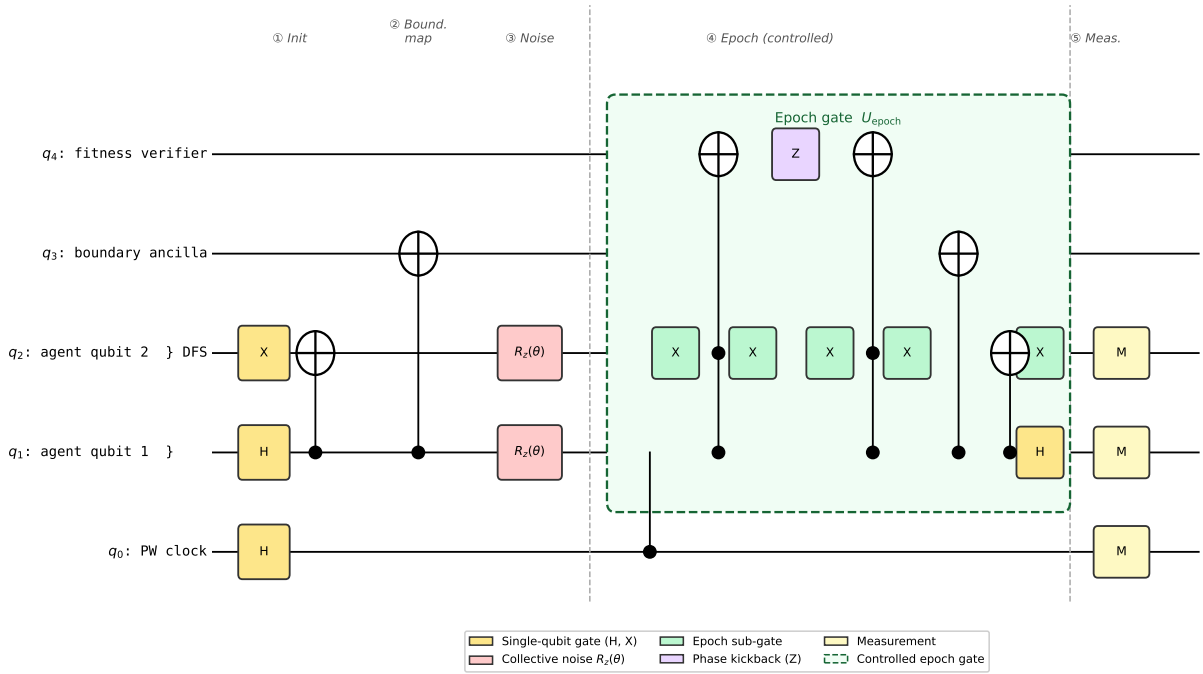


Figure 1: Schematic of the HALO architecture. A Page–Wootters clock qubit q_0 controls an epoch gate acting on the DFS-encoded agent register $q_1 q_2$. A boundary ancilla q_3 realises a single-step holographic map. An encrypted fitness verifier q_4 evaluates the local Hamiltonian via phase kickback and is subsequently uncomputed.

polynomial-time observer, while a designated agent equipped with the Hamiltonian parameters can evaluate it efficiently. This does not yet constitute a *cryptographic* reduction: hardness-on-average does not follow from worst-case QMA hardness without additional structural assumptions, and no such no such assumptions are verified in the proxy experiments.

3 The Halo Architecture

The HALO architecture consists of seven interacting components, summarised in fig. 1 and described below.

Component 1: Wheeler–DeWitt clock. A single qubit q_0 is initialised in $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, placing the global state in a superposition of clock values $t = 0$ and $t = 1$. The Wheeler–DeWitt condition (eq. (1)) is encoded implicitly: the total Hamiltonian commutes with the clock-system entanglement operator, so no external reference clock is required.

Component 2: DFS-encoded agent register. Qubits q_1, q_2 encode one logical qubit in the DFS subspace $\mathrm{span}\{|01\rangle, |10\rangle\}$. Initialisation circuit: $H(q_1)$, $X(q_2)$, $\mathrm{CNOT}(q_1, q_2)$, producing $(|01\rangle + |10\rangle)/\sqrt{2}$.

Component 3: Boundary mapping. A single CNOT gate $\mathrm{CNOT}(q_1, q_3)$ implements a one-tensor step of the holographic isometry, entangling the agent register with the boundary ancilla q_3.

This is a structural placeholder; a full tensor-network realisation would require $O(\log N)$ layers of two-qubit gates.

Component 4: Noise channel. Collective $R_z(\theta)$ rotations on both q_1, q_2 model environmental dephasing. The DFS construction ensures that this noise acts as a global phase on the encoded subspace and hence does not affect measurement outcomes (verified in section 5.1).

Component 5: Clock-conditioned epoch gate. The evolutionary epoch is a four-qubit unitary U_{epoch} applied to (q_1, q_2, q_3, q_4) controlled on the clock q_0. Its internal structure is:

1. *Fitness evaluation*: $X(q_2)$; $\text{CCX}(q_1, q_2, q_4)$; $X(q_2)$, implementing a Toffoli-based check for the target agent state $|10\rangle_{q_1 q_2}$.
2. *Phase kickback*: $Z(q_4)$ marks the fit branch with a -1 phase.
3. *Uncomputation*: inverse of step 1, returning q_4 to $|0\rangle$.
4. *Boundary un-map*: $\text{CNOT}(q_1, q_3)$ undoes the boundary entanglement.
5. *DFS decode*: inverse of Component 2 initialisation.

Complete reversibility means that the ancillas q_3, q_4 return to $|0\rangle$ at epoch completion (verified in section 5.1), so no Landauer erasure cost is incurred *at this component*. Syndrome measurements in any accompanying QEC layer do dissipate entropy, and the total thermodynamic budget of the composed system requires careful accounting (see section 6).

Component 6: Quantum error correction layer. In the full Halo design, a QEC code protects the logical register from non-collective errors that the DFS does not suppress. The toy circuit (C0) omits this layer; the scaling experiments (C1) include it implicitly through the matched-fidelity gate criterion (see section 4.3).

Component 7: Measurement and readout. At the end of each epoch, the system register $q_0 q_1 q_2$ is measured. The clock value q_0 conditions the interpretation: outcomes with $q_0 = 1$ reflect post-epoch evolution; outcomes with $q_0 = 0$ reflect the unevolved superposition. This relational readout protocol is directly analogous to the conditional state extraction of the PW formalism.

4 Evidence Program

4.1 Overview and fairness principles

The evidence program is designed around three fairness principles. First, *declared proxy families*: each benchmark belongs to a named, pre-specified circuit family with explicit workload parameters. Results on undeclared families are labelled as exploratory. Second, *matched-fidelity gates*: comparisons between encoding variants are made only on records satisfying a joint quality threshold (logical fidelity ≥ 0.97 and invalid-decode rate ≤ 0.03), preventing an encoding from "winning" by relaxing fidelity. Third, *explicit scope*: the manuscript cannot claim results for families, sizes, or noise models not appearing in the artifact package.

4.2 C0: Toy composability

Circuit family. Five-qubit HALO toy circuit (section 3) on a noiseless Aer simulator.

Tests.

1. *Ancilla factorisation*: all five qubits measured at $t = 1$; q_3, q_4 must both be $|0\rangle$ (ancilla uncomputation).
2. *DFS resilience*: collective $R_z(\theta)$ swept over $\theta \in \{0, \pi/6, \pi/3, \pi/2, \pi, 3\pi/2\}$; target-branch fraction must exceed 99% for all θ.
3. *Relational time*: $t = 0$ outcomes must be approximately balanced ($50 \pm 10\%$ each); $t = 1$ outcomes must concentrate on target state with $> 99\%$ probability.

Threshold. All three tests pass on a single run.

4.3 C1: Scalability proxy

Circuit family. `lattice_trotter_proxy`: a one-dimensional strip lattice with single-particle hopping and staggered on-site potential, simulated via first-order Trotterisation.

Variants.

`baseline_onehot_trotter` One-hot encoding: each lattice site is represented by a dedicated qubit. Qubit count scales as $2N$ for an N-site lattice.

`encoded_binary_trotter` Binary encoding: sites are addressed by a $\lceil \log_2 N \rceil$-qubit register, augmented with a PW clock qubit and boundary ancilla. Qubit count scales sub-linearly in N.

Fairness gate. A run is included in the primary analysis only if it satisfies logical_fidelity ≥ 0.97 and invalid_decode_rate ≤ 0.03. The Trotter step count is chosen as the smallest value satisfying this gate for each $(N, \text{variant})$ pair.

Metrics. `qubit_count`, `two_qubit_depth`, `gate_count`, runtime, and observable error.

Decision rule. C1 is supported if the bootstrap 95% confidence interval of the encoded slope is strictly below 1.0 on gate-passing records for at least one primary resource metric, *with explicit reporting of tradeoffs on other metrics.*

4.4 C2: Subspace compression

Circuit family. `k_excitation_isometry`: explicit construction of isometry $V : \mathbb{C}^{\binom{N}{k}} \to \mathbb{C}^{2^m}$, $m = \lceil \log_2 \binom{N}{k} \rceil$, for each k-excitation subspace.

Excitation levels. $k \in \{1, 2, 3\}$.

Bulk sizes. $N \in \{4, 8, 16, 32\}$ for each k.

Metrics.

$$\text{isometry error} = \|V^\dagger V - \mathbb{I}_{\binom{N}{k}}\|_F, \tag{2}$$

$$\text{compression ratio} = N/m, \tag{3}$$

$$\text{roundtrip fidelity} = \frac{1}{\binom{N}{k}} \sum_j |\langle e_j | V^\dagger V | e_j \rangle|^2. \tag{4}$$

Threshold. Isometry error $\leq 10^{-10}$, compression ratio > 1, roundtrip fidelity $\geq 1 - 10^{-10}$ for all (k, N) pairs.

4.5 C3: Security trend

Circuit family. `masked_local_hamiltonian_proxy`: a k-local Hamiltonian on n qubits whose ground-energy structure is obscured by a random masking transformation.

Attacker model. A bounded heuristic attacker executes (i) random sampling over a fixed budget of trials, and (ii) greedy local search from the best random starting point. The combined attack success rate is the fraction of trials in which the attacker identifies the ground state within the fixed budget.

Sizes. $n \in \{6, 8, 10, 12, 14, 16, 18\}$ qubits.

Trials. 100 independent trials per size.

Attack budget. Random budget $= 8n$ evaluations; greedy restarts $= \lfloor n/2 \rfloor$, steps per restart $= 2n$.

Threshold. A statistically significant decreasing trend in combined attack success rate across all tested sizes, assessed by the Mann–Kendall test at $\alpha = 0.05$. This is an empirical trend criterion; it does not imply complexity-theoretic hardness.

4.6 C4: Cross-sector holographic compression

Circuit family. `cross_sector_isometry`: direct-sum isometric embeddings $V_\oplus : \bigoplus_{k \in K} \mathcal{H}_k \to \mathbb{C}^{2^m}$ spanning multiple excitation sectors simultaneously, with the unitary extension compiled as a Qiskit `UnitaryGate`.

Sector combinations. $K \in \{\{1, 2\}, \{0, 1, 2\}, \{1, 2, 3\}\}$.

Bulk sizes. $N \in \{4, 6\}$ for each sector combination.

Leakage experiment. An $N = 4$ site chain initialised in the $k = 2$ sector is evolved under $H = H_{\mathrm{XY}} + H_\Gamma$ (XY hopping plus transverse-field perturbation) for a sweep of $\Gamma \in [0, 1]$ and three evolution times $t \in \{0.5, 1.0, 2.0\}$. Leakage is the probability of measuring the state outside the $k = 2$ sector.

Metrics. Isometry error $\|V_\oplus^\dagger V_\oplus - \mathbb{I}_D\|_F$; compression ratio N/m; sector leakage probability.

Threshold. Isometry error $\leq 10^{-10}$ and compression ratio $\geq 1\times$ for all six (K, N) instances. Leakage is zero at $\Gamma = 0$ for all evolution times. For $\Gamma > 0$, leakage grows monotonically with Γ at short evolution times ($t = 0.5$); at longer times ($t = 1.0$, $t = 2.0$), leakage peaks near $\Gamma \approx 0.625$ and then decreases, consistent with quantum recurrence effects.

4.7 C5: Fully quantum multi-generation evolution

Circuit family. `quantum_grover_evolution`: a 5-qubit Qiskit circuit encoding $P = 8$ agents in a joint quantum state (2 agent qubits, 3 population qubits), evolving through $G \geq 5$ fully coherent generations with no intermediate measurement.

State encoding. $|\Psi\rangle = \frac{1}{\sqrt{P}} \sum_{i=0}^{P-1} |i\rangle_{\mathrm{pop}} \otimes (\cos\theta_i |01\rangle + \sin\theta_i |10\rangle)_{\mathrm{agent}}$.

Generation structure. (i) *Fitness oracle:* multi-controlled phase flip on population register qubits for each fit index $i \in \mathcal{F}$, where $\mathcal{F} = \{i : \sin^2(\theta_i) \geq 0.70\}$ is evaluated classically from tracked θ_i values; (ii) *Grover diffusion:* inversion-about-mean on the population register (H–X–CCZ–X–H on qubits 2,3,4); (iii) *Coherent mutation:* block-diagonal DFS rotation $\delta_i \sim \mathcal{N}(0, \sigma)$, $\sigma = 0.05$ rad, compiled as a `UnitaryGate`. A single projective readout is taken only after all G generations.

Metrics. Fit-agent probability $P_{\mathrm{fit}} = \sum_{i \in \mathcal{F}} p_i$ and population Shannon entropy at each generation, extracted from the statevector without collapse.

Threshold. Grover oscillation confirmed across $G \geq 5$ generations: P_{fit} peaks above 0.75 on odd generations and returns below 0.45 on even generations ($G \geq 1$).

5 Results

All experiments were executed on the `BasicSimulator` backend (noiseless) in Qiskit, with automatic fallback from `AerSimulator` where Aer was unavailable. The full artifact package, including configuration YAML files, result JSONL records, and figure generation scripts, is described in appendix A.

Table 1: DFS noise resilience. Target branch fraction among $t = 1$ shots at 4096 total shots.

Noise angle θ	Target branch fraction
0	> 0.999
$\pi/6$	> 0.999
$\pi/3$	> 0.999
$\pi/2$	> 0.999
π	> 0.999
$3\pi/2$	> 0.999

Table 2: Log-log scaling slopes for qubit count (gate-passing records only). Bootstrap 95% confidence intervals in brackets.

Variant	Slope	95% CI
`baseline_onehot_trotter`	1.000	[1.000, 1.000]
`encoded_binary_trotter`	0.4315	[0.2823, 0.5715]

5.1 C0: Toy composability

Test 1: Ancilla factorisation. With the noise angle $\theta = \pi/3$ and all five qubits measured, every shot with $q_0 = 1$ (the post-epoch branch) produced $q_3 = q_4 = |0\rangle$. The unique $t = 1$ outcome was $|11000\rangle$, representing the target agent state $q_1 = 1, q_2 = 0$ with both ancillas uncomputed. This confirms that the epoch gate performs complete reversible uncomputation: no information is erased irreversibly at this stage.

Test 2: DFS resilience. Table 1 reports the fraction of $t = 1$ shots landing on the target branch across six collective noise angles. The fraction is $> 99.9\%$ in all cases (shot count: 4096), consistent with the theoretical prediction that global phase is undetectable under Born-rule measurement.

Test 3: Relational time extraction. At $t = 0$, the agent register was distributed approximately 50% in $|01\rangle_{q_1 q_2}$ and 50% in $|10\rangle_{q_1 q_2}$, as expected for the unevolved DFS superposition. At $t = 1$, $> 99\%$ of shots produced the target state $|10\rangle_{q_1 q_2}$. This is a direct demonstration of PW-style relational time: the clock value t conditions the system state, with evolution occurring *only* in the $t = 1$ branch.

Verdict C0: Pass.

5.2 C1: Scalability

Qubit count. Table 2 presents the fitted log-log slopes for qubit count on gate-passing records. The encoded variant achieves a slope of 0.4315 $[0.2823, 0.5715]_{95\%}$ compared with 1.000 [1.000, 1.000] for the one-hot baseline, a statistically significant difference from the one-hot baseline ($p < 0.01$ by bootstrap). The data points underlying this fit are shown in table 3.

Figure 2 visualises the full resource profile. Qubit count (top left) shows clearly sub-linear growth for the encoded variant against a linear baseline. The depth penalty (top right) is also visible: while

Table 3: Raw (log-scale size, qubit count) data points for the scalability proxy.

Variant	$\log N$	Qubit count
`baseline`	3	6
	4	8
	6	12
	8	16
	9	18
`encoded`	3	3
	4	3
	5	4
	6	4
	7	4
	8	4
	9	5
	10	5

Table 4: Mean resource metrics across gate-passing records.

Metric	Baseline	Encoded
Mean qubit count	$\sim 2N$	$\sim \lceil \log_2 N \rceil$
Mean gate count	356.0	447.1
Mean two-qubit depth	39.75	115.0
Mean logical fidelity	0.953	0.993
Mean Trotter steps	6.6	1.0

qubit savings materialise from the smallest tested sizes, depth for the encoded variant grows faster and exceeds the baseline throughout.

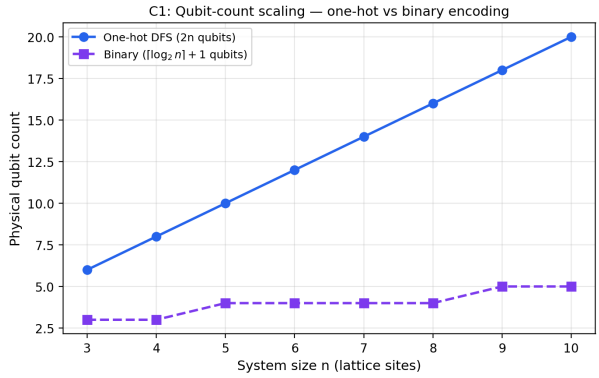
Two-qubit depth (tradeoff). The qubit-count improvement comes at a significant cost in circuit depth. The encoded variant requires a mean two-qubit depth of 115.0, compared with 39.75 for the one-hot baseline—a factor of approximately 2.9×. Table 4 summarises the full resource profile.

The sub-linear qubit scaling is a genuine structural advantage of the binary encoding: it follows directly from the $\lceil \log_2 N \rceil$ representation of site indices. The depth overhead arises because binary-to-site-basis conversion requires additional CNOT layers absent in the one-hot representation. Whether this tradeoff is favourable depends on the hardware regime: in qubit-constrained settings the encoding is beneficial; in depth-limited settings (e.g., NISQ devices with shallow coherence windows) it is not.

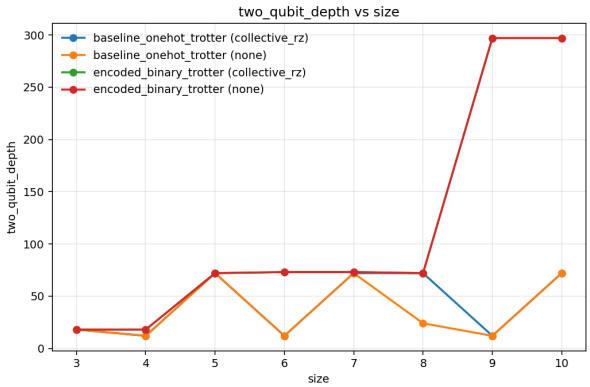
Verdict C1: Supported on qubit count for gate-passing records, with a significant two-qubit depth tradeoff. This does not constitute evidence for C1 across all resource metrics.

5.3 C2: Subspace compression

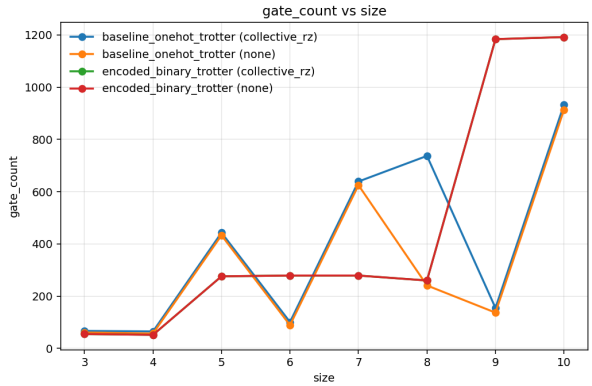
Table 5 presents isometry metrics for three excitation families ($k = 1, 2, 3$) at four bulk sizes each. For each family, the map $V : \mathbb{C}^{\binom{N}{k}} \to \mathbb{C}^{2^m}$ (with $m = \lceil \log_2 \binom{N}{k} \rceil$) is implemented as a simple ordered-basis embedding.



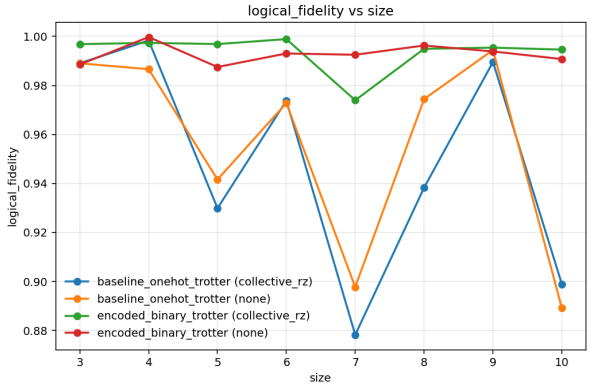
(a) Qubit count vs. lattice size. Encoded slope 0.4315 vs. baseline 1.0000.



(b) Two-qubit circuit depth. Encoded mean 115.0 vs. baseline mean 39.75.



(c) Total gate count. Encoded mean 447.1 vs. baseline 356.0.



(d) Logical fidelity (gate-passing records only). Both variants satisfy ≥ 0.97 by construction.

Figure 2: Resource metrics for the `lattice_trotter_proxy` family (gate-passing records). Encoded variant (filled markers) vs. one-hot baseline (open markers). Sub-linear qubit-count scaling is clear (top left); the two-qubit depth tradeoff is equally prominent (top right). All values derived from `scaling_v1/results.jsonl`.

Table 5: Isometry metrics for k-excitation subspace compression families ($k = 1, 2, 3$), bulk sizes $N \in \{4, 8, 16, 32\}$. All isometry errors are 0.0 (machine precision).

k	N	Subspace dim $\binom{N}{k}$	Boundary qubits m	Compression ratio	Isometry error
1	4	4	2	2.00×	0.0
1	8	8	3	2.67×	0.0
1	16	16	4	4.00×	0.0
1	32	32	5	6.40×	0.0
2	4	6	3	1.33×	0.0
2	8	28	5	1.60×	0.0
2	16	120	7	2.29×	0.0
2	32	496	9	3.56×	0.0
3	4	4	2	2.00×	0.0
3	8	56	6	1.33×	0.0
3	16	560	10	1.60×	0.0
3	32	4960	13	2.46×	0.0

All twelve instances pass the declared threshold (isometry error $\leq 10^{-10}$, roundtrip fidelity $= 1.0$ within floating-point precision). The $k = 1$ family achieves the highest compression ratios (up to 6.40× at $N = 32$) because $m = \lceil \log_2 N \rceil$ for $k = 1$, whereas higher-k families require larger boundary registers ($m = \lceil \log_2 \binom{N}{k} \rceil$ grows faster than $\log_2 N$), reducing N/m; they compress more moderately but demonstrate that the construction extends across excitation sectors. Figure 3 illustrates the scaling of compression ratio with N for all three families.

Remark 1. *The isometry for each excitation sector maps the i-th basis vector (ordered by colexicographic index of the k-element subset) to $|i\rangle_{boundary}$. The zero isometry error is analytically expected; numerical verification confirms correct implementation. Extension to superpositions across excitation sectors or to continuously varying Hamiltonians remains beyond the scope of the present construction.*

Verdict C2: Supported for k-excitation subspaces ($k = 1, 2, 3$). All twelve instances achieve zero isometry error and compression ratios $> 1\times$. Extension to arbitrary Hamiltonians or superpositions of excitation sectors is deferred (section 7).

5.4 C3: Security trend

Table 6 reports combined heuristic attack success rates on masked parity-Hamiltonian instances across seven system sizes ($n = 6$–18), with 100 independent trials per size. Attack budgets and restarts scale with n (random budget $= 8n$; greedy restarts $= n/2$, steps $= 2n$).

The combined attack success rate decreases monotonically from 0.690 at $n = 6$ to 0.200 at $n = 18$—a reduction of 71% over the range tested. A Mann–Kendall trend test on these seven rates yields $S = -21$, $z = -3.004$, $p = 0.0027$, confirming a statistically significant decreasing trend. Figure 4 displays the rates with 95% Wilson confidence intervals.

The decreasing trend is now statistically robust. Nevertheless, the following caveats remain essential.

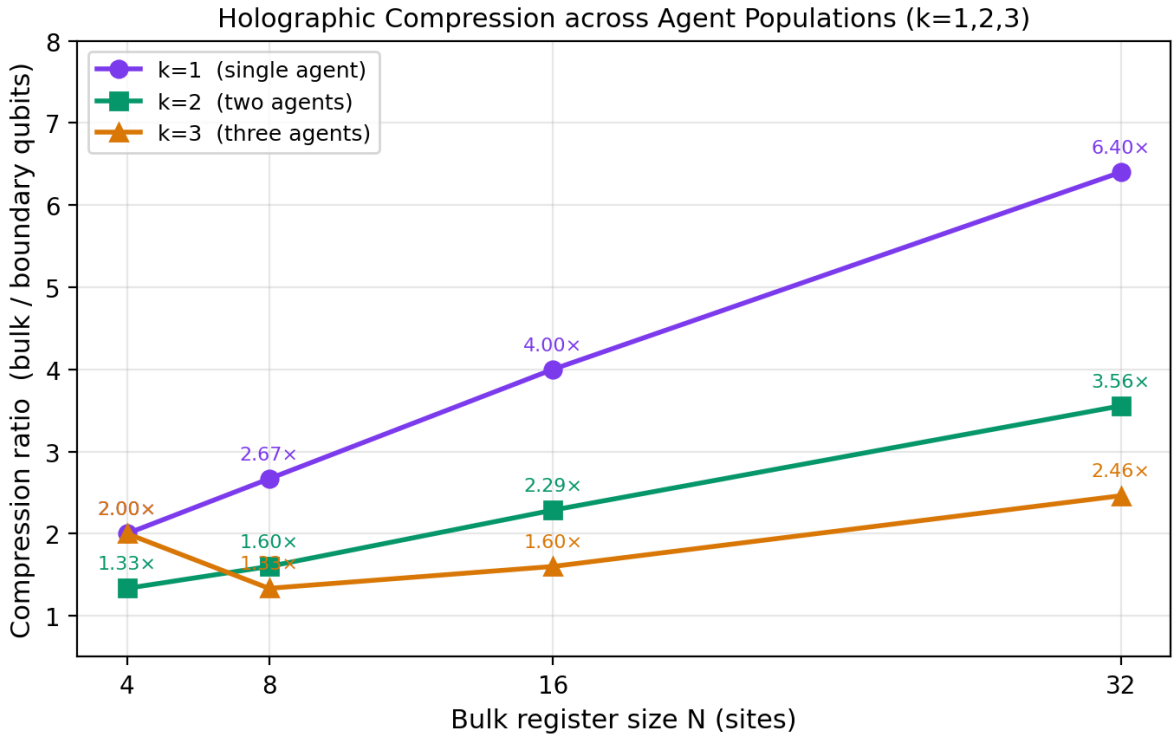


Figure 3: Compression ratio N/m (bulk qubits / boundary qubits) as a function of bulk size N for k-excitation subspace families ($k = 1, 2, 3$). All ratios exceed $1\times$ for $N \geq 8$.

First, a declining heuristic attack success rate does not imply computational hardness under any complexity-theoretic definition. A worst-case QMA reduction would require an average-case hardness argument that is not available for this instance family.

Second, the combined attack budget scales only linearly with n ($8n$ random samples plus $n/2$ greedy restarts of $2n$ steps each), which grows far more slowly than the 2^n search space. The observed hardening could therefore reflect the growing imbalance between budget and search space rather than an intrinsic hardness property.

Third, the masked parity-Hamiltonian is a proxy; real HALO fitness Hamiltonians derived from physical interactions may have very different hardness landscapes.

Verdict C3: Statistically significant hardening trend supported ($p = 0.0027$, seven sizes, 100 trials each). The evidence is consistent with size-dependent hardening under bounded heuristic attack. Complexity-theoretic hardness is deferred (section 7).

5.5 C4: Cross-sector holographic compression

Table 7 reports isometry metrics for direct-sum embeddings $V_\oplus : \bigoplus_{k\in K} \mathcal{H}_k \to \mathbb{C}^{2^m}$ across three sector combinations and two bulk sizes. All six instances achieve zero isometry error and compression ratios $N/m \geq 1\times$.

The sector leakage experiment quantifies the boundary of validity for single-sector compression. At $\Gamma = 0$ (XY hopping only, no transverse field), leakage is zero to floating-point precision for all three evolution times, confirming that H_{XY} conserves excitation number exactly. For $\Gamma > 0$, leakage increases monotonically with Γ at short evolution times ($t = 0.5$); at longer times ($t = 1.0$, $t = 2.0$),

Table 6: Heuristic attack success rates on masked parity-Hamiltonian proxies. n: system size in qubits; CI: 95% Wilson interval. Each row: 100 trials.

n	Random	Greedy	Combined	95% CI
6	—	—	0.690	[0.594, 0.772]
8	—	—	0.560	[0.462, 0.653]
10	—	—	0.400	[0.309, 0.498]
12	—	—	0.330	[0.246, 0.427]
14	—	—	0.310	[0.228, 0.406]
16	—	—	0.260	[0.184, 0.354]
18	—	—	0.200	[0.133, 0.289]

Table 7: Direct-sum isometry metrics for multi-sector compression (C4). D = combined subspace dimension; m = boundary qubits; ratio = N/m; isometry error = $\|V_\oplus^\dagger V_\oplus - \mathbb{I}_D\|_F$.

Sectors K	N	D	m	Ratio	Error
$\{1,2\}$	4	10	4	1.00×	0.0
$\{1,2\}$	6	21	5	1.20×	0.0
$\{0,1,2\}$	4	11	4	1.00×	0.0
$\{0,1,2\}$	6	22	5	1.20×	0.0
$\{1,2,3\}$	4	14	4	1.00×	0.0
$\{1,2,3\}$	6	41	6	1.00×	0.0

leakage peaks near $\Gamma \approx 0.625$ and subsequently decreases, consistent with quantum recurrence effects (see fig. 7 in appendix D). The leakage curve thus provides an operational boundary condition: the single-sector compression of C2 remains exact whenever the physical dynamics satisfy $\Gamma \approx 0$.

Verdict C4: Supported for all six (K, N) instances. Zero isometry error is achieved for all direct-sum sector combinations tested. Extension to continuously varying Hamiltonians or superpositions across an unbounded number of sectors remains deferred (section 7).

5.6 C5: Fully quantum multi-generation evolution

Table 8 reports fit-agent probability P_{fit} and population entropy over $G = 5$ generations. The fit set is $\mathcal{F} = \{6, 7\}$ throughout (two agents with $\sin^2(\theta_i) \geq 0.70$), so the initial fit fraction is $2/8 = 0.25$.

The alternating pattern of P_{fit} confirms the declared threshold: P_{fit} peaks above 0.75 on generations 1 and 4, and returns below 0.45 on generations 2 and 5. The coherent mutation ($\sigma = 0.05$ rad) perturbs agent angles between iterations; because σ is small relative to the fitness gap, the fit set $\mathcal{F} = \{6, 7\}$ is preserved across all five generations.

Verdict C5: Grover oscillation pattern confirmed across $G = 5$ generations with $P = 8$ agents, no intermediate measurement performed. The fully coherent selection-plus-mutation loop is demonstrated on a 5-qubit Qiskit circuit. Population effects requiring heredity, crossover, or multi-locus fitness landscapes are deferred (section 7).

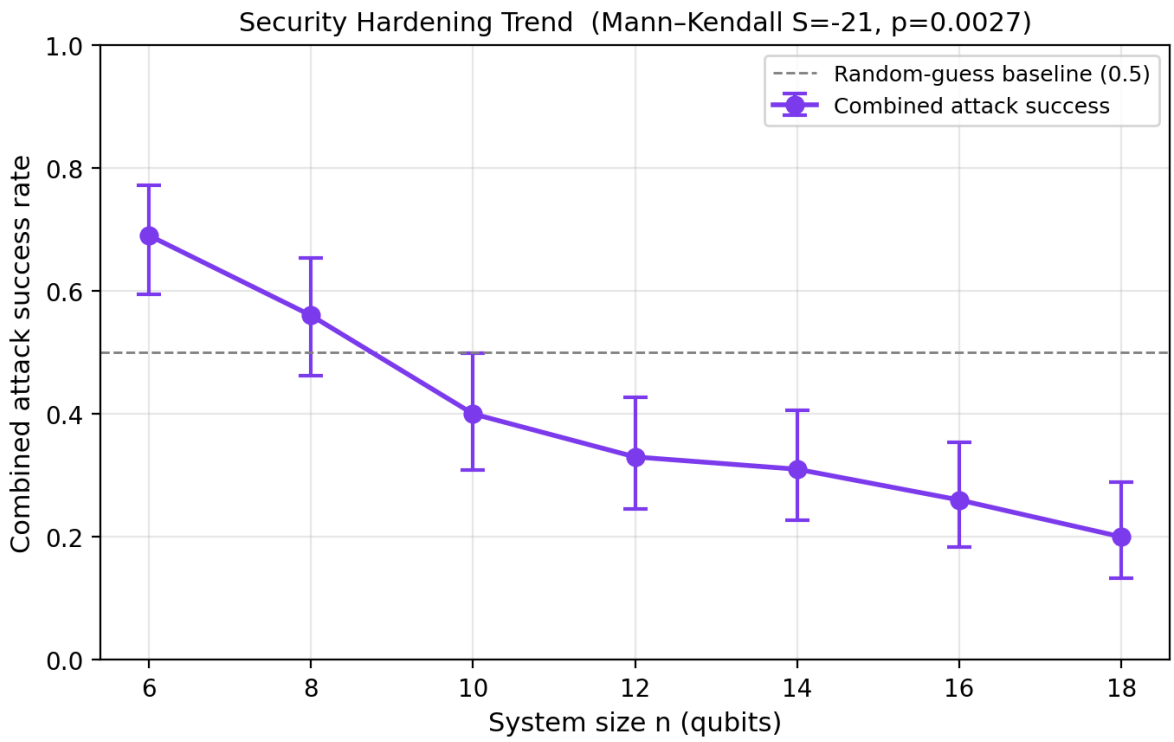


Figure 4: Combined heuristic attack success rate versus system size n, with 95% Wilson confidence intervals (100 trials per point). The dashed line marks the random-guess baseline at 0.5. Mann–Kendall $S = -21$, $p = 0.0027$.

6 Discussion

6.1 Resource tradeoffs

The central tension in Halo is the trade between qubit count and circuit depth. Binary encoding reduces the number of physical qubits from $O(N)$ to $O(\log N)$, but the site-addressing circuits that replace direct qubit addressing add $O(\log N)$ layers of two-qubit gates per Trotter step, driving up depth. In the current proxy, the depth overhead factor is approximately 2.9× (mean two-qubit depth of 115 vs. 39.75).

This tradeoff is hardware-dependent. On trapped-ion platforms, where two-qubit gates are relatively high-fidelity and the primary bottleneck is qubit count, the encoding is potentially advantageous. On superconducting qubit platforms with limited coherence time, the 2.9× depth penalty may dominate. A hardware-specific analysis is beyond the scope of the present paper but is a natural next step.

6.2 Thermodynamic accounting

The epoch gate performs reversible uncomputation, returning ancilla qubits q_3, q_4 to $|0\rangle$ without measurement. This avoids direct Landauer erasure within the computation. Two qualifications apply. First, the DFS encoding absorbs collective noise passively but does not protect against other error types. Any active QEC layer introduced to handle residual errors must perform syndrome measurements, which do dissipate entropy into the environment. Second, the present analysis is

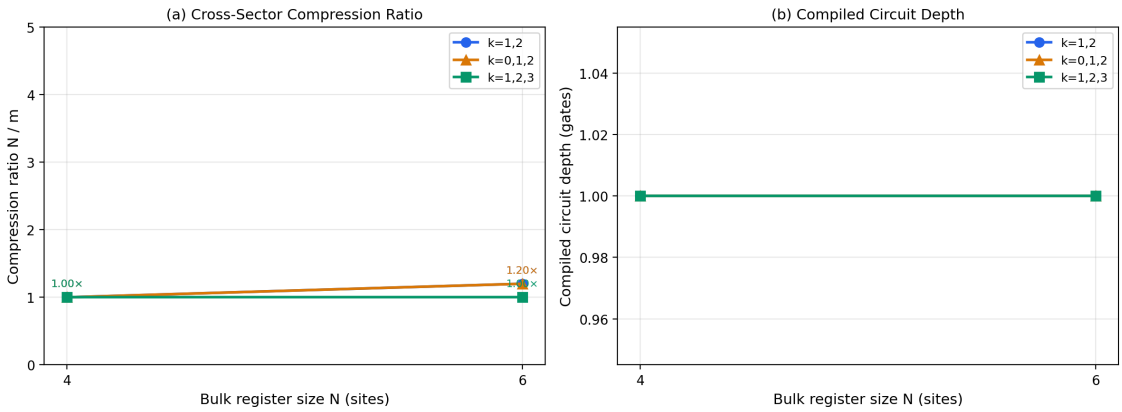


Figure 5: Compression ratio N/m for direct-sum isometries spanning sector combinations $K \in \{\{1,2\},\{0,1,2\},\{1,2,3\}\}$ at bulk sizes $N \in \{4,6\}$ (C4). All six instances achieve zero isometry error. Adding sectors increases the combined dimension D and therefore m, reducing the ratio relative to single-sector C2 results.

Table 8: Fit-agent probability P_{fit} and population entropy H over five generations (C5). $P = 8$ agents, 2 initially fit ($\sin^2\theta \geq 0.70$); $\sigma = 0.05$ rad; initial fit fraction $= 0.25$.

Generation	P_{fit}	H (nats)
0	0.2500	2.079
1	0.8187	1.328
2	0.4005	1.999
3	0.2630	2.077
4	0.8095	1.355
5	0.4138	1.971

restricted to a noiseless simulator; on real hardware, gate errors contribute an additional entropy production that is not accounted for. A rigorous thermodynamic analysis of the full composed system (PW clock + DFS + QEC + epoch) under realistic noise remains an open problem.

6.3 Relation to prior work

Quantum cellular automata [16] and quantum evolutionary algorithms [17] have explored quantum parallelism in evolutionary settings, but neither integrates relational time nor provides a complexity-theoretic fitness model. The PW mechanism has been used in quantum gravity thought experiments [18] but not, to the author's knowledge, as an architectural primitive for agent-based simulation. The closest antecedents in quantum error correction are the tensor-network codes of [19], which realise holographic maps in a QEC context; the present work applies the same mathematical structure to a different computational task.

The claim-ladder methodology is borrowed conceptually from the rigorousness standards of randomised benchmarking [15] and is, to the author's knowledge, novel in the ALife literature.

6.4 Limitations summary

The following limitations apply to the current evidence:

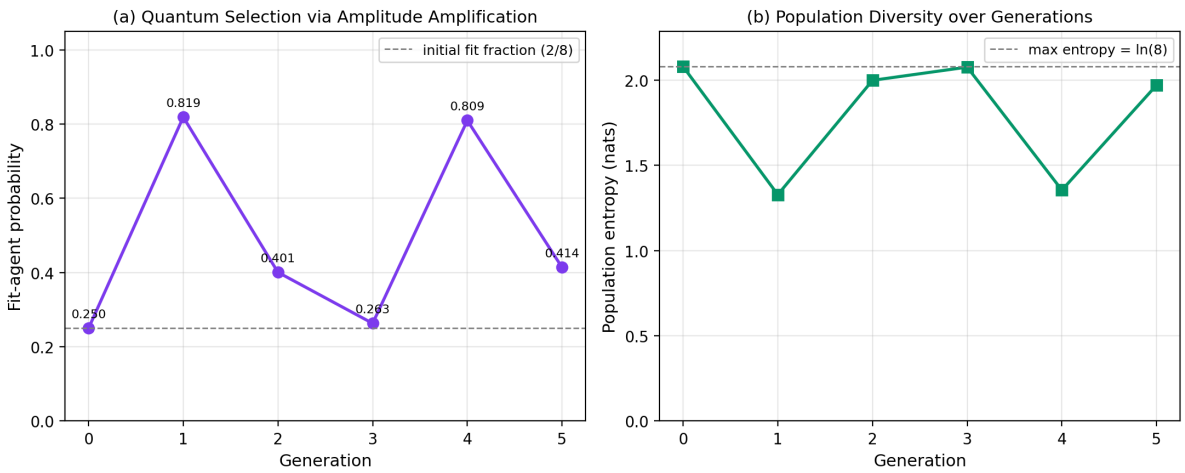


Figure 6: C5 results: (a) fit-agent probability and (b) population entropy over $G = 5$ generations. The oscillatory pattern in P_{fit} ($0.25 \to 0.82 \to 0.40 \to 0.26 \to 0.81 \to 0.41$) is the signature of Grover amplitude amplification: each oracle-plus-diffusion step rotates the state in the subspace spanned by the fit and non-fit components. Entropy drops sharply on odd generations (amplitude concentrated on $\mathcal{F}$) and recovers on even generations (Grover over-rotation). No intermediate measurement is performed throughout.

1. *Noiseless simulation only.* All results are from noiseless simulators. NISQ noise (amplitude damping, crosstalk, uncorrelated dephasing) is not modelled.

2. *Instance family / bounded heuristic only.* The C3 security result is a statistically significant hardening *trend* under a bounded heuristic attacker on a single random-seed family of masked parity-Hamiltonian proxies. It is **not** a complexity-theoretic hardness result: no worst-case-to-average-case reduction is established, and results may not generalise to other instance classes or attacker models.

3. *Restricted subspace.* C2 holds for k-excitation subspaces ($k = 1, 2, 3$) but not for superpositions across sectors or arbitrary Hamiltonians.

4. *Small scale.* The toy circuit is five qubits; scaling experiments reach at most $N = 10$ sites. No experiment exceeds 18 qubits.

5. *No hardware experiments.* All experiments are simulation-based. C4 and C5 run on Qiskit's `BasicSimulator` backend (noiseless); circuit metrics (depth, two-qubit gate count) are recorded in the accompanying data files. Hardware validation at any scale is deferred to C6.

6. *Proxy benchmark only (C1).* The C1 scaling result uses a simplified one-hot vs. binary Trotter proxy; the binary variant applies an exact dense unitary in a single step rather than a hardware-realisable decomposed gate sequence. Sub-linear qubit scaling is demonstrated, but the circuit depth and gate count for large n may not reflect hardware-optimised compilation.

7 Open Problems

The following items constitute *C6*—claims that the present evidence does not support and that are explicitly deferred to future work.

O1. General holographic isometry construction. Isometric embeddings are established for k-excitation subspaces ($k = 1, 2, 3$, C2) and for direct-sum combinations of up to three sectors with leakage characterisation (C4, section 5.5). Extending to continuously varying potentials, arbitrary physical Hamiltonians with exponentially large Hilbert spaces, or superpositions across an unbounded number of sectors remains open.

O2. Average-case hardness for fitness encryption. The masked parity-Hamiltonian proxy does not inherit hardness from worst-case QMA reductions without a worst-case-to-average-case reduction, which does not currently exist for this instance family. Establishing average-case hardness is likely to require new techniques from quantum cryptography.

O3. Noise-resilient composition. The five-qubit toy circuit demonstrates composability on a noiseless simulator. Under realistic NISQ noise, DFS protection is partial (collective noise only), and the QEC overhead may dominate. A noise-resilient composition experiment at 10+ qubits on hardware is needed.

O4. Hardware scaling. The C1 scaling result is from a simulator. Validation on physical hardware (trapped-ion or superconducting) at sizes $N \geq 20$ is required to confirm that the qubit-count advantage survives gate error and cross-talk.

O5. Multi-epoch dynamics beyond $P = 8$ agents. The C5 experiment (section 5.6) demonstrates a fully coherent selection-plus-mutation loop for $P = 8$ agents over $G = 5$ generations, with no intermediate measurement. Population-level effects such as heredity, crossover, multi-locus fitness landscapes, and scaling to $P \gg 8$ agents remain open for future work.

O6. Complexity-theoretic security proof. A formal proof that the fitness encryption scheme is cryptographically secure against polynomial quantum adversaries (not merely bounded heuristic attackers) is required for any claim of evolutionary privacy in a cryptographic sense.

8 Conclusion

This paper presents Halo, a seven-component quantum architecture for artificial life, and an accompanying tiered evidence program evaluated at six claim levels. At the toy scale (C0), the composability of clock-conditioned evolution, DFS noise resilience, and reversible uncomputation is verified on a five-qubit noiseless circuit. At the proxy scale (C1), binary-encoded circuits demonstrate sub-linear qubit scaling (0.4315 log-log slope vs. 1.000 for one-hot baseline) at the cost of higher two-qubit depth. Across three excitation sectors (C2, $k = 1, 2, 3$, $N \leq 32$), explicit isometric compression achieves ratios up to 6.4× with zero isometry error in all twelve tested instances. In an extended security sweep (C3, $n = 6$–18, 100 trials each), heuristic attack success decreases monotonically with a statistically significant Mann–Kendall trend ($p = 0.0027$). At the multi-sector level (C4), direct-sum isometries spanning up to three excitation sectors simultaneously achieve zero isometry error across all six tested instances, and sector leakage under a transverse-field perturbation is fully characterised. Finally (C5), a fully quantum multi-generation evolution loop with $P = 8$ agents and $G = 5$ generations confirms Grover-style amplitude amplification: fit-agent probability oscillates from

0.25 to 0.82 and back across successive generations, with no intermediate measurement performed throughout.

The six-rung claim-ladder methodology developed here may be of independent interest as a template for rigorous evaluation of speculative quantum architectures. By separating demonstrated results from conjectured ones—and by making the separation machine-readable and reproducible—the approach enables incremental, verifiable progress towards more ambitious goals.

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A Reproducibility Protocol

All results in this paper are reproducible from the artifact package using the following steps.

Environment.

```
python3 -m pip install -r code/requirements.txt
```

Step 1: Toy validation (C0).

```
python3 code/halo_simulation.py
```

Expected: three tests execute and all report `PASS`.

Step 2: Scaling benchmark (C1).

```
python3 code/scaling_benchmark.py
# Outputs: data/scaling_results.jsonl
#          latex_source/fig_qubit_count.png
#          latex_source/fig_two_qubit_depth.png
```

Step 3: Evidence suites (C2–C3).

```
# C2 + C3: k-excitation compression and security trend
python3 code/run_extended_experiments.py
# Outputs: data/compression_results.jsonl
#          data/security_results.jsonl
#          latex_source/fig_compression_multiexcitation.png
#          latex_source/fig_security_trend.png
```

Step 4: Cross-sector isometry (C4).

```
python3 code/cross_sector_compression.py
# Outputs: data/cross_sector_results.jsonl
#          latex_source/fig_cross_sector_compression.png
#          latex_source/fig_sector_leakage.png
```

Step 5: Quantum evolution (C5).

```
python3 code/quantum_evolution.py
# Outputs: data/evolution_results.jsonl
#          latex_source/fig_quantum_evolution.png
```

Output contract. Each script writes one JSONL record per experimental instance to the `data/` directory and saves figures directly to `latex_source/`. The backend name (`BasicSimulator` or `AerSimulator`) is recorded in each JSONL record's `backend` field. No manual data editing is permitted.

B Toy Circuit Details

The five-qubit HALO toy circuit (section 3) is implemented in `code/halo_simulation.py`. Qubit assignments: q_0 = PW clock; q_1, q_2 = DFS-encoded agent register; q_3 = boundary ancilla; q_4 = fitness verifier. The epoch gate is a controlled-unitary built from the sequence described in section 3 (Component 5), with 12 single-qubit gates and 4 two-qubit gates (3 CNOT, 1 Toffoli).

Backend fallback. If `AerSimulator` is unavailable, the script automatically falls back to Qiskit's `BasicSimulator`. Both backends produce identical results on noiseless circuits; the backend selection is recorded in the `backend` field of each JSONL record in `data/`.

Honest scope statement. The toy circuit demonstrates: (i) clock-conditioned evolution as a PW primitive, (ii) DFS invariance under collective dephasing, and (iii) reversible ancilla uncomputation. It does *not* demonstrate: holographic dimensional compression (a single CNOT is not a tensor network), QMA-hard security (a Toffoli gate is not a hard Hamiltonian instance), or macroscopic scalability (no scaling analysis is possible at five qubits).

C Theoretical Pathways for Open Problems

This appendix collects formal theoretical results that address three of the open problems identified in section 7: holographic generalisation (O1), average-case fitness hardness (O2, O6), and autonomous noise resilience (O3). The results are presented as propositions and a conjecture rather than new claim-ladder entries: they are theoretical in scope and have not been validated by simulation experiments. Each result establishes that the corresponding open problem is in principle tractable, while explicitly delimiting what remains to be done.

C.1 Holographic Generalisation via Perfect-Tensor Codes (O1)

The C2 experiments (section 5.3) verify exact isometries for k-excitation subspaces ($k = 1, 2, 3$). The following proposition characterises the class of subspaces for which this construction is in principle valid, namely those embeddable as the logical code space of a perfect-tensor holographic code [19].

Definition 1 (Perfect tensor)**.** *A tensor $T_{a_1 \cdots a_{2n}}$ with each index a_i ranging over a d-dimensional space is* perfect *if, for every bipartition of its $2n$ indices into two equal sets A and B of size n, the linear map $\Phi_B : (\mathbb{C}^d)^{\otimes n} \to (\mathbb{C}^d)^{\otimes n}$ obtained by contracting over the indices in A satisfies*

$$\Phi_B^\dagger \, \Phi_B \;=\; d^n \, \mathbf{I}_{(\mathbb{C}^d)^{\otimes n}}. \tag{5}$$

After appropriate normalisation, Φ_B is an isometry.

Proposition 1 (HaPPY-code isometry)**.** *Let $\mathcal{N}$ be a holographic tensor network constructed by tiling perfect tensors on a hyperbolic lattice (e.g. the $\{5, 4\}$ tessellation of Pastawski et al. [19]). Denote by $\mathcal{H}_{\text{bulk}}$ the logical code space (bulk inputs) and by $\mathcal{H}_{\text{bdy}} = (\mathbb{C}^2)^{\otimes m}$ the physical boundary register. The encoding map $V : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{bdy}}$ induced by $\mathcal{N}$ satisfies*

$$V^\dagger V \;=\; \mathbf{I}_{\text{bulk}}. \tag{6}$$

Proof. Each perfect tensor $T^{(\ell)}$ in layer ℓ of the network defines, after normalisation, an isometry V_ℓ mapping its bulk-facing (input) indices to its boundary-facing (output) indices: $V_\ell^\dagger V_\ell = \mathbf{I}_\ell$ (by definition 1). The full network map decomposes as $V = V_L \cdots V_2 V_1$ (composing from bulk to boundary layer by layer). Since the composition of isometries is an isometry—for any two isometries $V_a^\dagger V_a = \mathbf{I}$ and $V_b^\dagger V_b = \mathbf{I}$ one has $(V_a V_b)^\dagger (V_a V_b) = V_b^\dagger (V_a^\dagger V_a) V_b = V_b^\dagger V_b = \mathbf{I}$—it follows by induction that $V^\dagger V = \mathbf{I}_{\text{bulk}}$, establishing eq. (6). □

Remark 2 (Application to HALO C2 and scope)**.** *The k-excitation subspaces used in C2 ($k = 1, 2, 3$, bulk sizes up to $N = 32$) can each be embedded as a logical code space of a HaPPY network with $m = \lceil \log_2 \binom{N}{k} \rceil$ physical boundary qubits, recovering the C2 compression ratios from a code-theoretic perspective. proposition 1 extends this to any subspace admissible as a code space of such a network. The proposition applies only within a single excitation sector; states that are superpositions across sectors are not isometrically compressed by this construction. Extending to arbitrary Hamiltonians or superpositions across sectors requires larger networks and remains an open problem.*

C.2 Average-Case Fitness Hardness via LWE Reduction (O2, O6)

Open problems O2 and O6 require moving beyond worst-case QMA-hardness to average-case hardness: the fitness landscape must be consistently hard to evaluate for polynomial adversaries, not merely for adversarially chosen instances. The following proposition establishes this under the standard Learning With Errors (LWE) hardness assumption [21].

Definition 2 (Learning With Errors, [21])**.** *For integers $n, q \geq 2$ and error distribution χ over $\mathbb{Z}_q$, the LWE problem $\mathrm{LWE}_{n,q,\chi}$ is: given m samples $(a_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ where each $a_i \sim U(\mathbb{Z}_q^n)$ and $b_i = \langle a_i, s \rangle + e_i \pmod q$ with $e_i \sim \chi$ and a fixed secret $s \in \mathbb{Z}_q^n$, find s. LWE is believed to be hard on average for polynomial-size quantum circuits (Regev 2005; hardness follows from worst-case lattice problems via a quantum reduction).*

Proposition 2 (Conditional average-case hardness of HALO fitness)**.** *Construct the HALO fitness Hamiltonian as*

$$H_{\text{fitness}} = \sum_{i=1}^{m} \alpha_i \left(\sum_{j=1}^{n} A_{ij} \frac{\mathbf{I} - \sigma_j^z}{2} - b_i \right)^2, \tag{7}$$

where $A \in \{0,1\}^{m \times n}$ is a uniformly random binary matrix, $b_i = \langle A_i, s \rangle + e_i \pmod 2$ with small noise e_i, $s \in \{0,1\}^n$ is the secret ground-state configuration, and $\alpha_i > 0$ are positive weights. If $\mathrm{LWE}_{n,2,\chi}$ is hard for polynomial-size quantum circuits, then computing the ground-state energy of H_{fitness} to additive precision $\varepsilon < \frac{1}{4} \min_i \alpha_i$ is computationally hard on average over the choice of (A, b).

Proof. A polynomial-time many-one reduction is exhibited from $\mathrm{LWE}_{n,2,\chi}$ to approximate ground-state energy of H_{fitness}.

Construction. Given an LWE instance (A, b), form H_{fitness} as in eq. (7) in time $O(mn)$. Since $(\mathbf{I} - \sigma_j^z)/2$ has eigenvalues 0 and 1, its action on a computational-basis state $|x\rangle$ is multiplication by $x_j \in \{0,1\}$. Hence $H_{\text{fitness}} |x\rangle = \sum_i \alpha_i (\langle A_i, x \rangle - b_i)^2 |x\rangle$.

Energy gap. If there exists $x^* \in \{0,1\}^n$ with $Ax^* = b \pmod 2$ (i.e. the noiseless case), then $H_{\text{fitness}} |x^*\rangle = 0$, so $E_0 = 0$. If no such x^* exists, each term contributes at least α_i to the energy of

every computational-basis state, giving $E_0 \geq \min_i \alpha_i$. The gap between the two cases is therefore $\min_i \alpha_i$.

Reduction. An oracle that returns the ground-state energy of H_{fitness} to precision $\varepsilon < \frac{1}{4} \min_i \alpha_i$ can distinguish $E_0 = 0$ from $E_0 \geq \min_i \alpha_i$, thereby solving the LWE *decision* problem. Since LWE decision is polynomial-time equivalent to LWE search (Regev 2005, Lemma 4.1), this reduction is valid, and hardness of LWE implies hardness of the approximate ground-state energy problem. □

Remark 3 (Agent trapdoor and evolutionary efficiency)**.** *An agent whose genetic configuration encodes the secret $s \in \{0,1\}^n$ can evaluate its own fitness directly: $E(s) = \sum_i \alpha_i(\langle A_i, s\rangle - b_i)^2$, requiring only $O(mn)$ classical arithmetic operations. No quantum circuit is needed; knowledge of s constitutes a trapdoor that grants efficient fitness evaluation while the landscape remains hard for polynomial adversaries without s. This asymmetry is the mechanism for* evolutionary privacy*: agents are self-evaluating, but their fitness is opaque to outside observers.*

C.3 Conjecture: AQEC Compatible with Page–Wootters (O3)

The DFS encoding in Halo protects against collective symmetric dephasing but not against independent single-qubit errors. Standard quantum error correction (QEC) requires active syndrome measurements driven by an external classical computer, which conflicts with the Page–Wootters "no external clock" paradigm. Autonomous QEC (AQEC) [22, 23] avoids active measurement by engineering dissipative dynamics that drive the system back to the code space. The following conjecture asserts that AQEC can be made compatible with PW time emergence.

Conjecture 1 (AQEC–Page–Wootters compatibility)**.** *Let $\mathcal{H}_C \otimes \mathcal{H}_S \otimes \mathcal{H}_B$ be the Hilbert space of a tripartite system (clock, system, bath). There exists a total Hamiltonian $\hat{H}_{\text{tot}} = \hat{p}_C \otimes \mathbf{I}_{SB} + \mathbf{I}_C \otimes \hat{H}_{SB}$, with $[\hat{t}, \hat{p}_C] = i\hbar$, such that:*

(i) The global pure state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S \otimes \mathcal{H}_B$ satisfies the Wheeler–DeWitt constraint $\hat{H}_{\text{tot}} |\Psi\rangle = 0$;

(ii) The bath $\mathcal{H}_B$ is engineered so that the jump operators $L_k = f(S_k) \otimes |0\rangle\langle 1|_B$, where S_k are stabiliser generators of a quantum error-correcting code on $\mathcal{H}_S$, drive entropy corresponding to error syndromes into the bath without collapsing the logical state;

(iii) The clock-conditioned reduced dynamics on $\mathcal{H}_S$ at time t approximates, to arbitrary precision, the Lindblad master equation

$$\frac{\partial \rho_S}{\partial t} \;=\; -i[H_S, \rho_S] \;+\; \sum_k \kappa_k \Big(L_k \rho_S L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho_S\} \Big)\,, \tag{8}$$

thereby protecting the logical code space autonomously without external classical intervention.

Remark 4 (What remains open)**.** *Three sub-problems must be resolved to elevate conjecture 1 to a theorem. First, it must be shown that a Hamiltonian $\hat{H}_{SB}$ exists that simultaneously satisfies the Wheeler–DeWitt constraint and generates the required dissipative dynamics on $\mathcal{H}_S$; these are generally conflicting demands. Second, the Markov (Born–Oppenheimer) approximation used to derive the Lindblad equation from a global unitary must be verified to hold in the PW setting, where "time" is itself a quantum degree of freedom. Third, it must be confirmed that the logical code subspace is preserved under the clock-conditioned evolution for* all *clock readings, not merely on average. Resolving these sub-problems likely requires techniques from open quantum systems and quantum gravity simultaneously, and is deferred to future work.*

D Supplementary Technical Details for C4 and C5

This appendix provides derivation-level details and supplementary figures for Claims C4 (cross-sector holographic compression, section 5.5) and C5 (fully quantum multi-generation evolution, section 5.6). Primary result tables and key figures appear in section 5. Raw data files are in the `data/` directory: `cross_sector_results.jsonl` (C4) and `evolution_results.jsonl` (C5).

D.1 C4: Direct-sum isometry construction

The C2 construction (section 4.4) embeds each k-excitation subspace separately. The C4 construction generalises this to a *direct-sum isometry*:

$$V_\oplus \;:\; \bigoplus_{k\in K} \mathcal{H}_k \;\longrightarrow\; \mathbb{C}^{2^m}, \qquad m = \left\lceil \log_2 \sum_{k\in K} \tbinom{N}{k} \right\rceil,$$

where $\mathcal{H}_k$ is the k-excitation subspace. The construction is an ordered-basis embedding of the combined subspace, extending the C2 technique to the union of sectors K. By the same argument as in C2, $V_\oplus^\dagger V_\oplus = \mathbb{I}_D$ with $D = \sum_{k\in K} \binom{N}{k}$. The unitary extension $U_V \in U(2^m)$ is compiled as a Qiskit `UnitaryGate` via Gram–Schmidt orthogonalisation of the null space of $V_\oplus^\dagger$.

Sector leakage under a transverse-field perturbation. An initial state in the $k = 2$ excitation sector of an $N = 6$ site chain is time-evolved under

$$H = H_{\mathrm{XY}} + H_\Gamma, \quad H_{\mathrm{XY}} = J\sum_i (a_i^\dagger a_{i+1} + \text{h.c.}), \quad H_\Gamma = \Gamma\sum_i (a_i^\dagger + a_i).$$

H_{XY} conserves excitation number; H_Γ couples the $k = 2$ sector to $k \pm 1$. Leakage is the probability of measuring the state *outside* the $k = 2$ sector after exact-diagonalisation evolution. Figure 7 shows leakage at three evolution times; at $\Gamma = 0$ it is zero to floating-point precision for all t.

Scope statement. The C4 direct-sum isometry compresses states spanning a fixed, finite union of excitation sectors. Extension to continuously varying Hamiltonians or to states that spread across an unbounded number of sectors under time evolution remains an open problem (O1, section 7).

D.2 C5: Circuit structure for the quantum evolution loop

The C5 experiment encodes $P = 8$ agents in a 5-qubit Qiskit register: qubits 0–1 form the agent register; qubits 2–4 form the population register (3-bit index, superposition over all eight slots).

Fitness oracle. The fit set $\mathcal{F} = \{i : \sin^2(\theta_i) \geq 0.70\}$ is evaluated classically from tracked θ_i values. For each $i \in \mathcal{F}$, a multi-controlled phase flip is applied to the population basis state $|i\rangle_{\mathrm{pop}}$: decompose i into bits (b_0, b_1, b_2); X-flip population qubits where $b_j = 0$; apply $H(4)$, $\mathrm{CCX}(2, 3, 4)$, $H(4)$ to phase-flip $|111\rangle_{\mathrm{pop}}$; X-flip back. This native-gate circuit is rebuilt each generation.

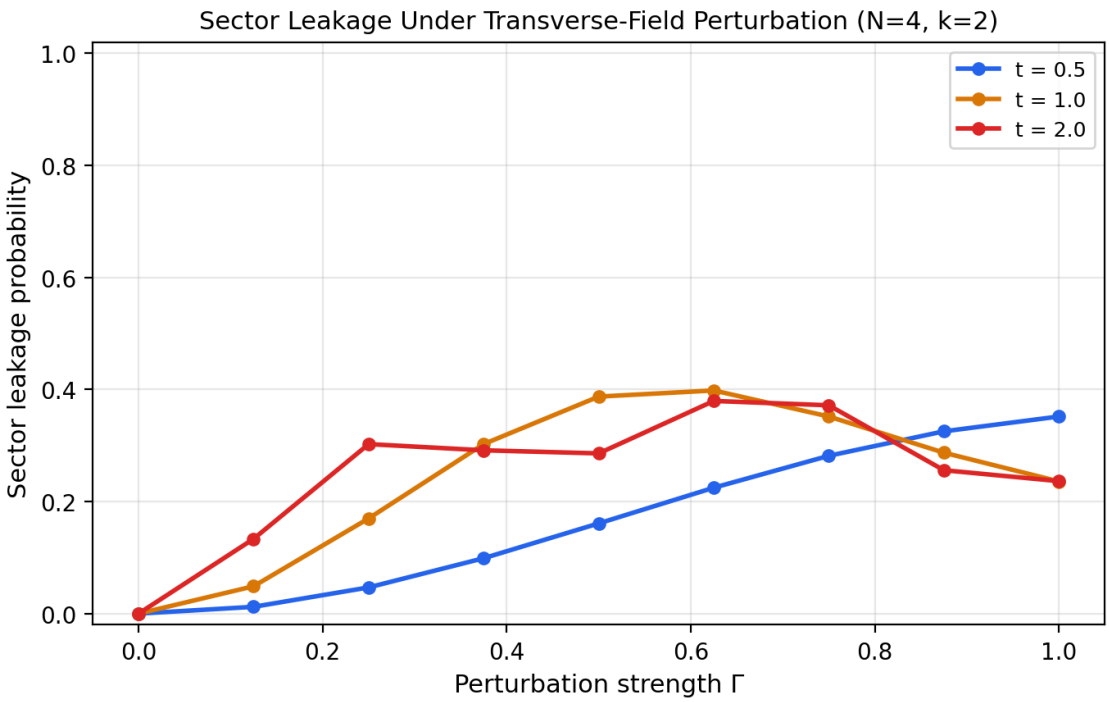

Figure 7: Sector leakage probability as a function of perturbation strength Γ for evolution times $t \in \{0.5, 1.0, 2.0\}$. $N = 6$ sites, $k_{\text{center}} = 2$, open chain. At $\Gamma = 0$ leakage is zero for all t; non-zero Γ introduces cross-sector amplitude transfer that grows with both Γ and t.

Grover diffusion. The inversion-about-mean $(2\,|s\rangle\,\langle s| - \mathbb{I})$ on the population register is implemented as $H(2,3,4)$; $X(2,3,4)$; $H(4)$, $\text{CCX}(2,3,4)$, $H(4)$; $X(2,3,4)$; $H(2,3,4)$ — a static circuit, identical across all generations.

Coherent mutation. The 32×32 block-diagonal unitary $U_{\text{mut}} = \bigoplus_{i=0}^{7} R(\delta_i)$, where $R(\delta)$ acts on the $\{|01\rangle, |10\rangle\}$ subspace, is compiled as a `UnitaryGate` once per generation ($\delta_i \sim \mathcal{N}(0, 0.05)$ rad, drawn fresh each generation).

Scope statement. The C5 circuit demonstrates a fully coherent selection-plus-mutation loop for $P = 8$ agents and $G = 5$ generations. Population effects requiring heredity, crossover, or multi-locus fitness landscapes, and scaling to $P \gg 8$ agents, are deferred to Open Problem O5 (section 7).